

Yes. We can hold Incident 02 exactly there and now expand **Incident 03 as one complete playable mission**.

Incident 03 is important because this is where the learner proves that the symmetric-security system can actually work **operationally**. Nothing malicious happens here. The learner needs a clean success baseline before Incident 04 can meaningfully break that trust.

Incident 03 — Complete the Symmetric Link

1. Core purpose

Incident 02 ended here:

```
ALICE
03_ACCESS_SCHEDULE.XLSX      ✓
Ciphertext                   ✓
Key A7                       ✓

BOB
Ciphertext                   ✓
Key A7                       ✗

CONTROLLED KEY-TRANSFER METHOD
Approved                     ✓

KEY RELEASE
Pending
```

So Incident 03 asks:

Before Phantom Q releases the shared secret, how do we make sure the person receiving it really is Bob and is authorised to receive it?

Then, once that is established:

Can Alice and Bob use the complete symmetric-security process successfully in a real operational task?

That is the whole incident.

2. What Incident 03 must NOT do

This incident must remain a **successful baseline mission**.

Do not introduce:

- unknown attackers;
- intercepted traffic;
- fraudulent payment instructions;
- compromised Key A7;
- suspicious IPs;
- public/private keys;
- signatures;
- Eve attacking the communication;
- R1 000;
- account 7394.

All of that belongs later.

Incident 03 needs the learner to finish thinking:

This works. I understand the process. I trust the workflow.

That trust is what Incident 04 will later challenge.

3. Starting state

The learner returns to the Cryptography Lab / Communications area.

The mission state is:

```
INCIDENT 03
COMPLETE THE SYMMETRIC LINK

ALICE
Original file      ✓
Ciphertext        ✓
Key A7            ✓

BOB
Ciphertext        ✓
Key A7            ✗

TRANSFER METHOD
Approved          ✓

NEXT REQUIREMENT
Verify Bob before releasing A7
```

Bob is remote.

Alice is present at HQ.

Eve may remain available as authorised Security Support, but she is not central to the transaction.

4. Incident 03 structure

The locked Incident 03 is **20 scenes**.

There are really two major halves:

PART A

Scenes 1-12

COMPLETE THE SYMMETRIC LINK

Identity → authorisation → key release → decrypt → verify

then:

PART B

Scenes 13-20

R100 OPERATIONAL PRACTICAL

Build algorithm → protect payment → send → decrypt → verify → process

The R100 practical must come **only after** the learner has successfully completed the file-transfer workflow.

PART A — COMPLETE THE SYMMETRIC LINK

Scene 1 — Bob Is Waiting

Bob's workstation shows:

PROTECTED FILE

03_ACCESS_SCHEDULE.XLSX

RECEIVED

✓

KEY A7

NOT AVAILABLE

×

DECRYPTION

LOCKED

The learner sees the approved key-transfer mechanism from Incident 02.

But:

```
RELEASE KEY A7  
LOCKED
```

Reason:

RECIPIENT NOT YET VERIFIED

This creates the next task naturally.

Scene 2 — Confirm Bob's Identity

The learner opens Bob's Phantom Q profile.

At Act I level we keep identity verification simple.

They may see:

```
PHANTOM Q USER  
  
Name/Role:  
Bob  
  
Unit:  
Authorised Receiver  
  
Current Session:  
Active  
  
Registered Identity:  
Available
```

The learner compares the active Bob session against the registered Bob identity.

Player action

CONFIRM IDENTITY

Result:

```
IDENTITY MATCH  
BOB ✓
```

This teaches:

Before giving a secret to somebody, first establish who they are.

Scene 3 — Authenticate Bob

Now we go one step deeper.

Identity says:

This profile belongs to Bob.

Authentication asks:

Is the current person/session able to prove that identity?

The learner runs the Phantom Q authentication check.

We don't need passwords, MFA internals or certificate theory yet.

Just:

```
REGISTERED BOB
    ↓
AUTHENTICATION CHECK
    ↓
CURRENT BOB SESSION
    ↓
MATCH ✓
```

Result:

```
BOB
AUTHENTICATED ✓
```

This is an important distinction without making it overly academic.

Scene 4 — Confirm Authorisation

Being Bob does not automatically mean Bob should receive every secret.

So the learner checks:

```
RESOURCE
03_ACCESS_SCHEDULE.XLSX
```

SECRET

KEY A7

REQUESTED RECIPIENT

Bob

ROLE

Authorised receiver

ACCESS

APPROVED ✓

This introduces another subtle but important idea:

IDENTITY

Who are you?

AUTHENTICATION

Can you prove it?

AUTHORISATION

Are you allowed to do this?

We don't need a long lecture. The learner experiences all three.

Scene 5 — Release Key A7

Only now does the controlled-transfer mechanism unlock.

BOB IDENTITY ✓

BOB AUTHENTICATED ✓

BOB AUTHORISED ✓

Player selects:

RELEASE KEY A7

The controlled method delivers the shared secret.

Result:

ALICE

KEY A7 ✓

BOB

KEY A7 ✓

This is the first time both parties hold the same shared secret.

Scene 6 — Confirm the Shared Secret

The learner compares the two key states.

```
ALICE
A7

BOB
A7
```

The system establishes:

MATCHING SHARED SECRET

This is where the term **symmetric** should really become operationally clear.

```
SAME KEY
ALICE ↔ BOB
```

Not because a definition told them so, but because they have physically established the same key at both ends.

Scene 7 — Bob Decrypts the File

Bob now has:

```
CIPHERTEXT  ✓
KEY A7      ✓
```

The learner activates:

DECRYPT

Process:

```
CIPHERTEXT
  +
KEY A7
  ↓
DECRYPT
  ↓
PLAINTEXT
```

The file becomes readable again:

```
03_ACCESS_SCHEDULE.XLSX
```

This completes the symmetric cycle:

```
PLAINTEXT
↓
ENCRYPT WITH A7
↓
CIPHERTEXT
↓
TRANSFER
↓
DECRYPT WITH A7
↓
PLAINTEXT
```

Scene 8 — Recover the Original Information

Now Bob can open the spreadsheet.

But **readable does not automatically mean correct**.

That distinction matters.

The learner has to ask:

Did Bob recover Alice's actual original information?

So Bob's recovered file is held beside Alice's source.

Scene 9 — Verify Bob's File Against Alice's Original

The player compares:

```
ALICE ORIGINAL
03_ACCESS_SCHEDULE.XLSX

↑ COMPARE

BOB RECOVERED
03_ACCESS_SCHEDULE.XLSX
```


Result:

MATCH ✓

CONTENT

UNCHANGED ✓

This is the proper end-to-end verification.

The learner has now achieved:

ORIGINAL SOURCE

↓

PROTECT

↓

TRANSFER

↓

RECOVER

↓

COMPARE

↓

MATCH

This should feel like a significant success.

Scene 10 — Training Replay: Wrong Key

Now we deliberately run an authorised **training replay**.

Not an attacker.

The system gives Bob:

KEY B4

instead of:

KEY A7

Player attempts:

DECRYPT

Result:

DECRYPTION FAILED ✕

or unreadable output.

The game should explain through the result:

```
ALICE USED  
A7  
  
BOB USED  
B4  
  
KEYS DO NOT MATCH
```

No generic:

Wrong key.

The process itself shows why it failed.

Scene 11 — Matching A7 Works

Reset training replay.

Bob uses:

KEY A7

Result:

```
DECRYPTION SUCCESSFUL ✓  
  
RECOVERED FILE  
MATCHES ORIGINAL ✓
```

The learner has now tested both conditions:

```
DIFFERENT KEY  
FAIL  
  
SAME KEY  
SUCCESS
```

That strengthens understanding without additional theory.

Scene 12 — Rebuild the Complete Symmetric Workflow

Now remove most guidance.

The learner receives workflow blocks:

```
CONFIRM SOURCE
VERIFY RECIPIENT
AUTHORISE RECIPIENT
ENCRYPT
SEND
RELEASE KEY
DECRYPT
COMPARE
ACCEPT / INVESTIGATE
```

They must construct the process.

A suitable operational flow is:

```
CONFIRM ORIGINAL
  ↓
ENCRYPT WITH A7
  ↓
SEND CIPHERTEXT
  ↓
VERIFY BOB
  ↓
AUTHORISE BOB
  ↓
RELEASE A7
  ↓
BOB DECRYPTS
  ↓
COMPARE WITH ORIGINAL
  ↓
MATCH?
  ↙      ↘
YES      NO
  ↓      ↓
ACCEPT  INVESTIGATE
```

The exact interface can simplify this, but the logic must remain.

Result:

SYMMETRIC LINK COMPLETE

At this point Incident 03 could technically end.

But now we test whether the learner can **apply the system**, not merely reproduce the file exercise.

PART B — R100 OPERATIONAL PRACTICAL

This is where the payment comes in.

It is **not present throughout the incident**.

It appears only after the symmetric workflow is working.

Scene 13 — Alice Has One More Request

Alice approaches the trainee again.

She has a fictional Phantom Q training payment.

The exact locked data:

PHANTOM Q TRAINING PAYMENT

Beneficiary:

PQ Equipment Services

Account:

.... 4821

Amount:

R100.00

Reference:

PQ-TEST-03

Transaction ID:

PQ-TX-003

This is deliberately a small fictional training payment.

Alice's instruction is the source of truth.

Scene 14 — Confirm Alice's Original Instruction

Before doing anything, the learner inspects the request.

ALICE ORIGINAL

Beneficiary	PQ Equipment Services
Account	 4821
Amount	R100.00
Reference	PQ-TEST-03
Transaction	PQ-TX-003

Player action

CONFIRM ORIGINAL

This creates the baseline that later matters enormously in Incident 04.

The learner now knows:

R100
4821

is the legitimate instruction.

Scene 15 — Payment Information Requires Protection

The learner attempts ordinary send.

System responds:

PAYMENT INFORMATION
PROTECTION REQUIRED

This mirrors Incident 02 but without reteaching everything.

The learner already knows why.

The key question now becomes:

Can you build the process yourself?

Scene 16 — Build the Payment Algorithm

This scene is crucial.

We do not simply give them an **Encrypt Payment** button.

The learner receives operational blocks.

For the locked practical:

```
CONFIRM ORIGINAL
↓
ENCRYPT WITH KEY A7
↓
SEND CIPHERTEXT
↓
BOB DECRYPTS WITH KEY A7
↓
VERIFY MATCH?
  ↙      ↘
YES      NO
  ↓      ↓
PROCESS STOP / INVESTIGATE
R100
```

The learner physically assembles this workflow.

This reinforces algorithmic thinking from Incident 01.

The game is not merely teaching cryptography. It is teaching **process design**.

Scene 17 — Test the Payment Algorithm

Before using the actual training instruction, the learner runs a simulation.

The test should check things such as:

SOURCE CONFIRMED	✓
ENCRYPTION STEP	✓
SHARED KEY A7	✓
BOB DECRYPTION	✓
MATCH CHECK	✓
FAILURE BRANCH	✓

If they build:

```
DECRYPT
↓
SEND
↓
ENCRYPT
```

the simulation produces:

PROCESS ORDER INVALID

If they omit verification:

UNVERIFIED PAYMENT OUTPUT

PROCESS CANNOT CONTINUE

Again, no generic wrong answer.

The algorithm must make operational sense.

Scene 18 — Run the Algorithm on the R100 Instruction

Now the actual training instruction enters the process.

ALICE ORIGINAL

R100

4821

↓

ENCRYPT WITH A7

↓

CIPHERTEXT

The ciphertext is transferred to Bob.

No attack.

No interference.

No suspicious activity.

This is a clean successful transaction.

Scene 19 — Bob Decrypts and Verifies

Bob receives the protected instruction.

He already holds the same:

KEY A7

So:

CIPHERTEXT

+

A7

↓

DECRYPT

Result:

Amount	R100.00
Account	•••• 4821
Reference	PQ-TEST-03

Then:

BOB RECEIVED
↑
ALICE ORIGINAL

Result:

MATCH ✓

This is extremely important.

Do not introduce public/private terminology here.

Do not introduce signatures.

Do not change A7.

It is the same **shared symmetric Key A7**.

Scene 20 — Process the R100 Payment

Only after successful verification does the process allow:

PROCESS PAYMENT

Result:

PQ-TX-003

R100.00


```
↓
PQ Equipment Services
.... 4821

STATUS
COMPLETED ✓
```

Alice confirms receipt / training payment success.

Bob sees:

```
PAYMENT SUCCESSFUL
```

The learner has completed the first full operational security workflow.

5. Why the R100 mission matters so much

It does several things simultaneously.

First, it proves that the learner can apply the concept outside the file exercise.

Second, it establishes a **known-good financial baseline**:

```
R100
↓
4821
↓
SUCCESS
```

Third, it establishes Alice and Bob's behavioural baseline:

```
ALICE
Creates legitimate instruction

BOB
Receives protected instruction
Decrypts
Verifies
Processes
```

Fourth, it creates trust in the system.

The learner should leave Incident 03 believing:

When Alice sends a protected instruction using A7 and Bob decrypts and

verifies it, the workflow is trustworthy.

Incident 04 needs that confidence.

6. Player actions throughout Incident 03

The learner should physically:

1. open the pending A7 release;
2. inspect Bob's registered identity;
3. confirm Bob's identity;
4. authenticate Bob;
5. check Bob's authorisation;
6. release Key A7;
7. compare Alice and Bob's key state;
8. activate Bob's decryption;
9. recover the file;
10. compare Bob's recovered file to Alice's original;
11. run the wrong-key B4 replay;
12. see decryption fail;
13. rerun with A7;
14. rebuild the full symmetric workflow;
15. receive Alice's R100 training-payment request;
16. inspect the original payment instruction;
17. confirm `R100 / 4821`;
18. construct the payment algorithm;
19. test the algorithm;
20. run the real R100 training instruction;
21. encrypt with A7;
22. send ciphertext;
23. decrypt at Bob with A7;
24. compare against Alice original;
25. approve/process the R100 payment;
26. observe successful completion.

That is a substantial operational mission.

7. Information flow

We need to keep tight control over what the learner knows.

Start

Bob has ciphertext.
Bob needs A7.
Controlled delivery exists.

After identity verification

Current receiver is Bob ✓

After authentication

Current Bob session is authenticated ✓

After authorisation

Bob is allowed to receive A7 ✓

After release

Alice and Bob both hold A7.

After decryption

The same shared secret can recover
the protected information.

After verification

Bob's recovered file matches Alice's original.

After wrong-key replay

A different key does not produce
the intended recovered information.

After R100 practical

The complete symmetric workflow
can protect and successfully process
a real operational instruction.

That is the knowledge boundary.

The learner does **not** know that A7 will later become compromised.

8. Character roles

Trainee

Incident 03 is where the trainee becomes much more operational.

They move from:

“I know how encryption works.”

to:

“I can establish and execute a controlled secure workflow.”

That is a significant clearance/capability progression.

Alice

Alice remains:

Origin / Protector

She owns:

- original file;
- Key A7 initially;

- legitimate payment instruction.

Her important role in the R100 practical is:

ALICE ORIGINAL

R100 / 4821

That source-of-truth relationship must be visually memorable because Incident 04 returns to it.

Bob

Bob remains:

Receiver / Verifier

Incident 03 is especially important for Bob.

He:

- proves identity;
- proves authorisation;
- receives A7;
- decrypts;
- compares;
- processes the legitimate payment.

This establishes him as a **competent receiver following the process correctly**.

That way Incident 04 does not feel like Bob simply made a stupid mistake.

Eve

Eve remains:

Security Support / Monitoring

She can be visible in the Cryptography Lab or monitoring environment.

But she has no attack role.

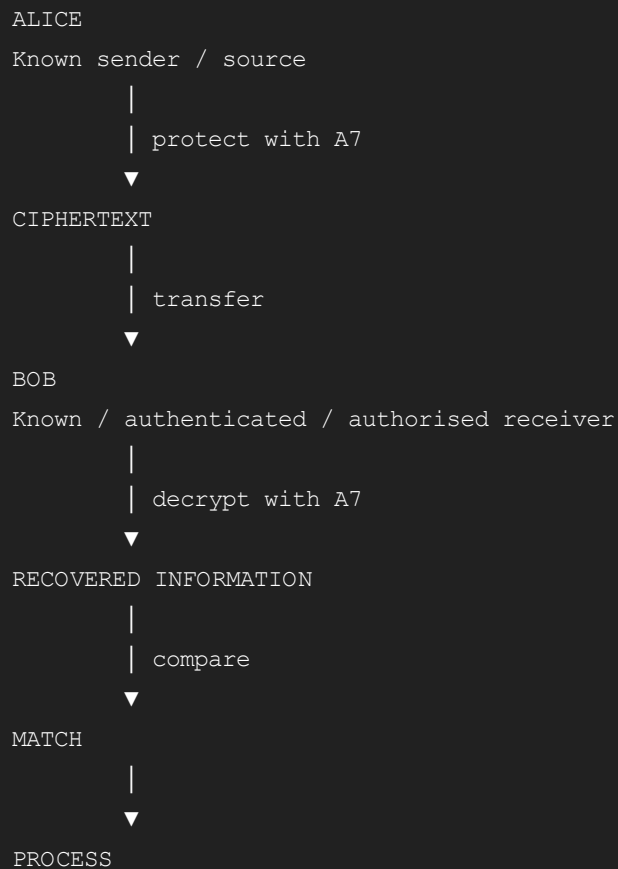
No intercept-resend.

No secret stealing.

No suspicious villain treatment.

9. The operational trust model established

By the end, the learner believes this:



That becomes the trusted model Incident 04 exploits.

10. Game mechanics

Incident 03 should use several mechanics rather than becoming repetitive.

Identity verification

Compare current session and registered identity.

Authorisation gate

The key-release object remains physically locked until required checks pass.

Key-state visualisation

ALICE

A7 ✓

BOB

A7 ✕

becomes:

ALICE

A7 ✓

BOB

A7 ✓

Decryption workbench

CIPHERTEXT + A7 → PLAINTEXT

Comparison mechanic

Original vs recovered information.

Wrong-key replay

A7 vs B4.

Algorithm builder

The learner creates the R100 payment workflow.

This should be the largest player-driven mechanic in the second half.

11. Success and failure behaviour

We should retain our campaign principle.

Release A7 before identity verification

KEY RELEASE BLOCKED

RECIPIENT IDENTITY

NOT VERIFIED

Identity confirmed but authorisation not checked

RELEASE BLOCKED

Identity does not automatically
grant access to this secret.

This teaches the difference between identity and permission.

Use B4

DECRYPTION FAILED

ALICE ENCRYPTION KEY
A7

BOB DECRYPTION KEY
B4

KEY MISMATCH

Skip file comparison

System:

RECOVERY NOT VERIFIED

Readable information alone
does not prove it matches
the original source.

Excellent connection back to integrity.

Build a payment algorithm without a match check

For example:

CONFIRM

↓

ENCRYPT

↓

SEND

↓
DECRYPT
↓
PROCESS

System should show:

UNVERIFIED PROCESS

No decision confirms that Bob's recovered instruction matches Alice's original.

So the learner adds:

VERIFY MATCH?

Process when comparison fails

In the training algorithm simulation:

MATCH?
NO
↓
PROCESS

should cause:

UNSAFE LOGIC

A failed verification must not continue to payment processing.

Then the learner must route:

NO
↓
STOP / INVESTIGATE

This becomes hugely important later.

12. What real concepts sit underneath the gameplay

Player experience	Real concept
Confirm Bob	Identity
Authenticate Bob	Authentication
Check permission	Authorisation
Release A7	Controlled key distribution
Alice A7 = Bob A7	Shared symmetric secret
Ciphertext + A7	Symmetric decryption
Compare recovered file	Integrity verification
B4 fails	Key matching
Build workflow	Algorithmic/process reasoning
Confirm payment source	Source of truth
Encrypt payment information	Data protection
Verify before processing	Transaction validation
R100 succeeds	Operational security baseline

This gives Incident 03 significant learning depth without dumping terminology.

13. What Incident 03 deliberately does not teach

Do not introduce:

- attacker interception;
- compromised shared secret;
- authenticity limitations of shared-key possession;
- public/private keys;
- digital signatures;
- IP addresses;
- networking;
- relays;
- S1;

- QKD.

The learner needs to believe the current architecture is **working correctly**.

14. Persistent outputs

Incident 03 creates several objects that become extremely important later.

Key A7 state

ALICE

A7

BOB

A7

Successful symmetric workflow

Confirm

Protect

Send

Recover

Verify

Process

Payment baseline

PQ-TX-003

Original amount:

R100

Original account:

.... 4821

Result:

SUCCESS

This should be saved as a persistent historical transaction.

Algorithm

The learner's approved payment algorithm becomes a reusable object.

That's important.

Incident 04 should say essentially:

Use your existing approved workflow.

Not make them rebuild it.

15. Incident 03 success condition

The learner succeeds when:

BOB IDENTITY
CONFIRMED ✓

BOB AUTHENTICATION
PASSED ✓

BOB AUTHORISATION
CONFIRMED ✓

A7 RELEASE
COMPLETE ✓

ALICE KEY
A7 ✓

BOB KEY
A7 ✓

FILE DECRYPTION
SUCCESS ✓

FILE VERIFICATION
MATCH ✓

WRONG-KEY REPLAY
UNDERSTOOD ✓

SYMMETRIC WORKFLOW
COMPLETE ✓

PAYMENT ALGORITHM
VALID ✓

PQ-TX-003
R100 → 4821 ✓

16. Incident 03 end-state

At the end the screen should communicate success, not suspicion.

INCIDENT 03

SYMMETRIC LINK
OPERATIONAL ✓

ALICE
A7 ACTIVE ✓

BOB
A7 ACTIVE ✓

FILE TRANSFER
PROTECTED ✓

DECRYPTION
VERIFIED ✓

PAYMENT
PQ-TX-003

R100 → •••• 4821
COMPLETED ✓

SECURITY STATUS
NORMAL

This is the last clean baseline before the campaign escalates.

17. The subtle hook into Incident 04

We should **not foreshadow too aggressively**.

The learner has just succeeded.

Maybe Alice simply says:

“Good. I have another payment request.”

That's enough.

The learner thinks:

I know how to do this now.

Then Incident 04 begins.

That is stronger than showing ominous warnings.

18. Why Incident 04 becomes powerful

Because Incident 03 established:

ALICE ORIGINAL

R100 / 4821

↓

A7

↓

BOB RECEIVED

R100 / 4821

↓

VERIFY

↓

PAYMENT SUCCESS

Then Incident 04 repeats the same trusted workflow but ends with:

ALICE ORIGINAL

R100 / 4821

↓

???

↓

BOB RECEIVED

R1 000 / 7394

The learner immediately understands the severity because they personally built and successfully tested the baseline.

Incident 03 master flow

```
INCIDENT 02 COMPLETE
↓
BOB HAS CIPHERTEXT
BUT NO A7
↓
VERIFY BOB IDENTITY
↓
AUTHENTICATE BOB
↓
CHECK AUTHORISATION
↓
RELEASE A7
↓
ALICE A7 = BOB A7
↓
BOB DECRYPTS FILE
↓
RECOVER PLAINTEXT
↓
COMPARE WITH ALICE ORIGINAL
↓
MATCH ✓
↓
TRAINING REPLAY
B4 → FAIL
↓
A7 → SUCCESS
↓
REBUILD COMPLETE
SYMMETRIC WORKFLOW
↓
SYMMETRIC LINK COMPLETE
↓
ALICE INTRODUCES
R100 TRAINING PAYMENT
↓
CONFIRM ORIGINAL
R100 / 4821
↓
PAYMENT INFORMATION
REQUIRES PROTECTION
↓
LEARNER BUILDS PAYMENT ALGORITHM
```

```
↓
TEST ALGORITHM
↓
RUN R100 INSTRUCTION
↓
ENCRYPT WITH A7
↓
SEND CIPHERTEXT
↓
BOB DECRYPTS WITH A7
↓
VERIFY AGAINST ALICE ORIGINAL
↓
MATCH ✓
↓
PROCESS PAYMENT
↓
PQ-TX-003
R100 → 4821
SUCCESS ✓
↓
TRUSTED SYMMETRIC BASELINE
ESTABLISHED
↓
INCIDENT 04
```

This is the complete **Incident 03 design specification** and preserves the critical rule:
Incident 03 succeeds cleanly; Incident 04 is where that trusted workflow is exploited.